

MILLAU, France

WMO: 075580

Lat: 44.1186N

Lon: 3.0194E

Elev: 2362

StdP: 13.48

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.9	(c) 23.4	(d) 9.6	(e) 9.8	(f) 26.9	(g) 14.1	(h) 12.2	(i) 29.0	(j) 36.6	(k) 38.5	(l) 32.9	(m) 36.4	(n) 11.4	(o) 310	(p) 0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.1	(c) 87.3	(d) 65.0	(e) 84.0	(f) 64.3	(g) 80.9	(h) 63.4	(i) 68.0	(j) 80.6	(k) 66.4	(l) 78.8	(m) 65.0	(n) 76.7	(o) 10.6	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	97.5	72.3	62.5	92.2	70.9	60.9	87.2	69.5	33.6	80.4	32.3	78.8	31.2	76.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 30.0	(b) 26.1	(c) 23.1	(e) 17.2	(f) 92.3	(g) 4.8	(h) 3.2	(i) 13.7	(j) 94.6	(k) 10.9	(l) 96.5	(m) 8.2	(n) 98.3	(o) 4.7	(p) 100.6		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			15.4	70.8	4.4	1.7	12.3	72.0	9.7	73.0	7.2	74.0	4.1	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	37.9	38.9	44.7	49.2	55.9	63.9	67.8	67.5	60.5	54.2	44.3	39.5
	DBStd	12.52	6.32	7.31	6.44	6.32	6.36	6.95	6.08	6.27	6.01	6.35	6.80	6.78
	HDD50	1542	377	316	185	90	19	1	0	0	3	35	191	325
	HDD65	5065	841	732	629	475	287	105	44	46	159	336	621	790
	CDD50	2311	0	4	22	65	203	417	553	543	318	165	20	1
	CDD65	358	0	0	0	6	71	131	124	25	1	0	0	0
	CDH74	2904	0	0	0	6	54	573	1073	1018	173	7	0	0
	CDH80	837	0	0	0	0	4	147	321	338	27	0	0	0
(14) Wind	WSAvg	10.7	10.5	11.3	11.9	12.1	11.2	10.5	10.5	9.7	9.6	10.1	10.9	10.1
(15) Precipitation	PrecAvg	35.7	2.8	2.3	2.2	3.5	3.4	2.4	2.0	2.4	3.7	3.8	4.4	2.8
	PrecMax	53.3	9.6	4.8	5.5	8.0	8.0	8.3	4.8	5.0	9.5	7.6	8.1	6.1
	PrecMin	26.3	0.0	0.1	0.0	0.5	1.2	0.7	0.4	0.5	0.6	1.8	1.2	0.3
	PrecStd	6.2	2.0	1.4	1.5	2.0	1.7	1.5	1.1	1.2	2.3	1.5	2.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	62.1	68.5	73.3	79.5	89.0	90.4	92.6	83.4	73.7	61.7	54.5
		MCWB	46.9	48.1	50.7	56.4	61.3	65.9	65.4	65.4	62.9	58.5	54.0	47.4
	2%	DB	50.3	55.7	63.2	68.4	74.7	83.8	86.7	87.9	78.7	68.9	58.2	51.8
		MCWB	46.5	45.9	49.6	53.6	59.2	64.9	64.6	65.3	61.0	57.9	53.2	47.8
	5%	DB	48.2	51.0	58.5	64.2	71.0	80.2	83.7	83.8	75.0	65.6	55.7	50.1
		MCWB	45.3	44.2	47.1	51.8	57.5	63.4	64.3	64.1	59.9	57.2	51.9	46.9
	10%	DB	46.2	47.8	54.4	60.1	67.3	76.8	80.5	80.1	71.1	62.6	53.4	48.3
		MCWB	43.9	43.5	45.7	49.8	55.7	62.3	63.4	63.3	59.2	56.7	50.4	45.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	50.4	53.3	57.6	63.8	70.2	69.9	69.5	66.4	62.4	56.4	50.8
		MCDB	51.4	58.4	63.0	69.8	75.5	82.0	82.6	82.5	77.2	67.6	59.7	52.1
	2%	WB	47.5	47.4	50.7	55.0	60.9	67.2	67.7	67.5	64.1	60.5	54.2	49.1
		MCDB	49.4	52.3	59.8	66.0	71.5	79.3	81.0	79.9	72.6	64.8	57.1	50.7
	5%	WB	45.7	45.7	48.9	52.8	58.6	65.1	66.2	66.2	62.4	59.1	52.6	47.6
		MCDB	47.6	49.3	55.9	61.8	68.2	76.9	79.1	78.2	70.0	63.1	55.1	49.3
	10%	WB	44.1	44.0	47.1	50.8	56.8	63.2	64.6	64.8	60.7	57.6	50.8	45.9
		MCDB	46.0	47.3	52.5	58.0	65.2	74.5	76.5	76.5	67.8	61.7	53.0	47.8
(35) Mean Daily Temperature Range	MDBR		9.7	12.2	14.6	15.5	16.0	18.0	19.1	19.0	16.7	12.9	10.3	9.7
	5% DB	MCDBR	12.2	18.0	21.9	22.1	21.4	22.9	24.3	24.5	22.4	17.2	13.2	11.8
		MCWBR	8.6	11.1	11.7	11.1	10.9	9.9	9.4	9.1	9.5	9.2	8.8	8.4
	5% WB	MCDBR	9.9	15.0	18.1	19.8	19.3	21.1	20.8	20.5	18.1	12.9	10.8	9.0
		MCWBR	7.9	10.1	10.5	10.6	10.8	10.1	9.2	9.0	9.0	7.7	7.8	7.3
(40) Clear-Sky Solar Irradiance	taub		0.286	0.301	0.337	0.368	0.376	0.379	0.373	0.372	0.349	0.339	0.308	0.284
	taud		2.517	2.478	2.402	2.315	2.338	2.357	2.371	2.383	2.435	2.473	2.512	2.526
	Ebn at Noon		271	285	286	284	284	282	283	279	278	266	259	260
	Edh at Noon		21	26	33	39	39	38	37	36	31	26	22	19
(44) All-Sky Solar Radiation	RadAvg		475	764	1131	1476	1752	2067	2065	1795	1387	872	529	423
	RadStd		55	90	114	169	154	148	144	114	92	90	66	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page